

Case Study 13 — Marcus–Hush Width

Classical Marcus–Hush broadening width from reorganization energy

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1 Case Study 13 — Marcus–Hush Width

Replace the fixed broadening sigma with the classical Marcus–Hush Gaussian width derived from a reorganization energy and temperature:

$$\sigma = \sqrt{2 \lambda k_B T}.$$

At $\lambda = 0.30$ eV and $T = 298.15$ K this gives $\sigma = 0.12416$ eV (FWHM = 0.29234 eV), which is narrower than the default $\sigma = 0.30$ eV.

The user-selectable knobs are:

- CLI: `--sigma-mode marcus-hush, --reorganization-ev FLOAT, --temperature-k FLOAT`
- API: form fields `sigma_mode=marcus-hush, reorganization_ev=FLOAT, temperature_k=FLOAT`

Physics note. This is the classical high-temperature Marcus–Hush limit. For organic vibrational baths (~ 1400 cm⁻¹), $\hbar\omega \gg k_B T$, so the result is a first-order estimate rather than a quantitative bandshape. For quantitative work use vibronic spectra via the experimental/tabular input branch (see [Case Study 15](#)).

1.1 What you will produce

Folder	By	Contents
output/cli/tables/	CLI	Result tables (CSV / JSON / XLSX)
output/cli/plots/	CLI	TIFF plots per sample, light source, and (sample, light) pair
output/api/	API	Same <code>tables/</code> and <code>plots/</code> tree, extracted from the ZIP
output/api/analysis_outputs.zip	API	Raw ZIP response

Light sources used: **AM1.5G** + **LED-B4**.

1.2 1. Prerequisites

If you have never run any case study before, follow [Case Study 01 §1 Prerequisites](#) once to install the package, the `requests` library, and either the Python or Docker option for the HTTP API. Every case study uses the same environment.

1.3 2. Inputs

The case-study `input.json` lists 2 Gaussian TD-DFT outputs. The manifest uses relative paths that point at the bundled data under `../input/files/`, so no files are copied or duplicated.

1.4 3. Run with the CLI

```
overlap-calculator analyze --input case_studies/13_marcus_hush_width/input.json --out
↪ case_studies/13_marcus_hush_width/output/cli --default-light-sources AM15G,LEDB4 --sigma-mode
↪ marcus-hush --reorganization-ev 0.30 --temperature-k 298.15 --plot-dpi 400
```

After the run the `sigma_ev` column in `results.*` shows the resolved Marcus–Hush sigma (≈ 0.12416 eV for the defaults above). The `sigma_mode` column shows `marcus-hush`.

1.5 4. Run with the HTTP API

```
curl -X POST http://localhost:8000/analyze -F "files=@input/files/slurm-2829207.out" -F
↪ "files=@input/files/slurm-2829212.out" -F "default_light_sources=AM15G,LEDB4" -F
↪ "sigma_mode=marcus-hush" -F "reorganization_ev=0.30" -F "temperature_k=298.15" -F "plot_dpi=400"
↪ --output case_studies/13_marcus_hush_width/output/api/analysis_outputs.zip
```

1.6 5. Drive both interfaces from one Python script

```
python case_studies/13_marcus_hush_width/submit.py
```

1.7 6. Expected output layout

```
case_studies/13_marcus_hush_width/output/
+-- cli/
|   +-- run_manifest.json
|   +-- tables/results.{csv,json,xlsx}          (2 samples x 2 light sources = 4 rows)
|   +-- tables/results_timings.{csv,json,xlsx}
|   +-- tables/descriptor_summary.{csv,xlsx}
|   +-- tables/ranking_by_light_source__<metric>.{csv,json,xlsx}
|   +-- tables/ranking_by_sample__<metric>.{csv,json,xlsx}
|   +-- plots/
+-- api/
    +-- analysis_outputs.zip
    +-- tables/...
    +-- plots/...
```

1.8 7. What to compare

- Compare `gaussian_molar_absorptivity_max` against the default $\sigma = 0.30$ eV run: the narrower Marcus–Hush sigma produces taller, narrower peaks.
- Re-run with `--reorganization-ev 0.60` to see how a larger reorganization energy widens the band.

1.9 8. Related case studies

- [Case Study 08 — Broadening Sigma](#)
- [Case Study 12 — Frequency-Resolved Prefactor](#)
- [Case Study 14 — Calibration Block](#)
- [Case Study 15 — Vibronic Tabular Branch](#)